

When Simple Wins: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

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ABSTRACT

Predictive modeling in low-data bioactivity regimes frequently succumbs to overfitting, silent data leakage, and unwarranted architectural complexity. To isolate the drivers of predictive fidelity, we executed a leak-free benchmarking framework evaluating classical machine learning ensembles against Directed Message Passing Neural Networks (ChemProp D-MPNN) for the inhibition of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $N = 756$). Subjecting four molecular representation spaces to a rigorous 25-fold nested cross-validation protocol revealed that explicit inductive bias supersedes algorithmic depth. Classical stacking ensembles trained exclusively on 2D Morgan fingerprints achieved optimal predictive performance ($R^2 = 0.7417 \pm 0.0493$, MAE = 0.4841). This lightweight topological approach statistically matched the performance of computationally intensive 2D/3D hybrid descriptors ($R^2 = 0.7371$, $p = 0.7605$) and strictly outperformed the D-MPNN ($R^2 = 0.6802 \pm 0.0595$, MAE = 0.5353), confirming that data scarcity restricts end-to-end representation learning. Deploying the optimized classical architecture to screen FDA-approved therapeutics prioritized high-potency antifolates and high-efficiency scaffolds, including triamterene (ligand efficiency = 0.49) and pyrimethamine (ligand efficiency = 0.52). Orthogonal validation via AutoDock Vina molecular docking resolved ligand-based screening discordances, filtering out structure-based false positives (e.g., bisacodyl) and confirming high-affinity consensus leads, most notably Pralatrexate (docking energy: -9.50 kcal/mol). In data-constrained QSAR environments, the fixed structural priors of 2D fingerprints act as a natural regularizer, providing a computationally efficient paradigm for target-specific drug repurposing that renders deep graph representations and 3D conformational overhead obsolete.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1. Introduction

1.1. Predictive Modeling in Low-Data Bioactivity Regimes

Predictive modeling in quantitative structure-activity relationships (QSAR) operates under strict constraints when data is scarce [?]. Under these conditions, simpler interpretable approaches—such as multivariate linear regression or classical tree-based ensembles—frequently match or exceed the prediction reliability of complex artificial neural networks ^{??}. These parsimonious models maintain higher bias and lower variance. They resist the extreme sensitivity to random fluctuations that plagues overparameterized algorithms, particularly when supplied with robust, quantum-chemically derived descriptors [?].

Small datasets inherently invite overfitting. This bottleneck worsens exponentially when the dimensionality of molecular descriptors exceeds the available sample size ^{??}. To circumvent data sparsity, contemporary

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workflows frequently deploy data augmentation (DAU) or synthetic sample generation to artificially expand the training manifold ^{??}. However, these interventions demand substantial computational overhead ^{??}. More critically, improper DAU application obscures the true predictive capacity of the base algorithms. When augmentation precedes dataset partitioning (the “augment-then-split” fallacy), it violates independent and identically distributed (i.i.d.) assumptions. Models memorize synthetic artifacts and latent data dependencies rather than learning genuine physicochemical rules, yielding artificially inflated performance metrics ^{???}.

Evaluating complex architectures in these regimes requires absolute methodological isolation. Hyperparameter tuning on small samples with large parameter spaces inevitably leads to model selection bias [?]. Consequently, assessing machine learning architectures in isolation provides zero evidence of real-world utility unless the model selection procedure itself undergoes rigorous validation. Double (or nested) cross-validation becomes mandatory to decouple model building from performance estimation ^{??}. Standard k-fold cross-validation systematically overestimates explorative capabilities ^{??}, and generative deep learning struggles to extrapolate within sparse, discrete chemical spaces like Simplified Molecular-Input Line-Entry System (SMILES) strings [?]. To preserve methodological integrity, any generative or augmentative process must remain strictly confined to the training folds after establishing an independent partition ^{??}.

To safely harness overparameterized models [?] without succumbing to the overfitting trap, empirical safeguards against chance correlation are non-negotiable. Techniques such as y-randomization (response shuffling) explicitly verify that the algorithm maps true chemical relationships rather than memorizing random noise [?]. Although frequently recommended in literature, its empirical execution is routinely omitted in modern deep learning QSAR studies; enforcing this protocol is therefore essential to definitively rule out chance correlation before advancing candidates to virtual screening. When descriptors undergo meticulous curation and pass these randomization checks, machine learning effectively guides rational design—successfully discriminating potent bioactives, such as Amyloid- β aggregation inhibitors, from decoys [?].

Alternatively, transfer learning bypasses data scarcity by adapting base networks optimized on massive source domains [?]. By enforcing reduced learning rates during fine-tuning [?] and prioritizing efficient convolutional architectures over heavier Transformer models [?], researchers can navigate class imbalance and model selection bias in low-resource settings.

1.2. Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies on domain-engineered descriptors and molecular fingerprints [?]. Representations such as Morgan (ECFP4) encode structural topologies, while physicochemical and 3D descriptors capture specific molecular interactions deterministically without requiring massive training corpora ^{???}. Embedding theoretical chemistry directly into the model ensures high interpretability and algorithmic transparency [?]. However, accumulating these handcrafted features generates highly dimensional, redundant matrices ^{??}. In restricted datasets, this redundancy amplifies computational costs and exacerbates the feature selection bottleneck ^{??}.

End-to-end representation learning bypasses manual feature selection. Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—autonomously map discrete molecular topologies into continuous latent spaces to model complex, nonlinear interactions ^{???}. Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES [?]. Yet, learning representations in discrete chemical spaces remains intrinsically fragile. Altering a single character in a SMILES string can irreparably invalidate the entire molecular structure [?].

The utility of deep learning complexity in low-data regimes remains highly contested. These algorithms require dense data manifolds, rendering representation learning ineffective when ground-truth labels are scarce [?]. To circumvent this limitation in *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) inhibition, recent studies deploy highly complex pipelines. [?] employed Gaussian noise-based data augmentation and deep neural network (DNN) ensembles to stabilize training on a restricted dataset, elevating their baseline R^2 from 0.75 to 0.85. This intervention prompts a critical evaluation: is synthetic data generation strictly necessary to achieve predictive reliability? Parsimonious classical ensembles frequently match or exceed the predictive fidelity of deep architectures in data-scarce scenarios, avoiding the catastrophic overfitting inherent to unregularized overparameterization ^{??????}.

Traditional skepticism toward nonlinear models in limited-data scenarios is evolving ?. Properly regularized nonlinear algorithms, hybridized with structured linear priors, achieve competitive results comparable to robust classical methods ?. Transfer learning serves as a primary catalyst for this shift. Fine-tuning pretrained networks like ChemBERTa allows researchers to extract accurate representation learning predictions using only dozens of target data points ?.

1.3. Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage—the unintended introduction of test data information into the training process—is a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings ?. In low-data QSAR modeling, this vulnerability frequently manifests across two primary vectors: preprocessing and model selection. When transformations like feature scaling, imputation, or variance-based filtering are applied globally across an entire dataset, the algorithm gains illicit knowledge of the test distribution. This violation of the independent and identically distributed (i.i.d.) assumption yields spuriously high performance estimates ?.

Similarly, traditional internal validation often suffers from model selection bias. If a validation set continuously influences hyperparameter tuning, it loses its statistical independence. Highly parameterized architectures, such as deep neural networks, are particularly prone to exploiting this overlap. They adapt to noise and produce deceptively optimistic metrics that severely underestimate true generalization errors ?. To preserve methodological integrity, model selection and preprocessing must operate as strictly encapsulated components of the fitting process. Insufficient reporting of these partitioning mechanics often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage ?.

1.4. Objectives and Principal Contributions

This study bridges the gap between stringent statistical validation and advanced representation learning to accelerate targeted drug discovery in low-data regimes. Our principal contributions are defined as follows:

- **Methodological Reproducibility and Leak-Free Validation:** Implementation of a fully transparent, reproducible Python pipeline built upon a rigorous 5×5 repeated nested cross-validation protocol (25 outer evaluations). This architecture ensures absolute statistical independence between hyperparameter optimization and model evaluation, eradicating model selection bias and data leakage.
- **Rigorous Statistical Inference in QSAR:** Introduction of the Nadeau-Bengio variance correction to molecular machine learning evaluations. By explicitly accounting for training set overlap across repeated folds, this framework resolves the pervasive issue of pseudoreplication in standard cross-validation. This methodological contribution enables mathematically sound hypothesis testing between architectures, preventing the Type I error inflation that frequently confounds model selection in cheminformatics.
- **Architectural Benchmarking in Data-Limited Regimes:** A comprehensive empirical comparison evaluating the predictive reliability of traditional domain-engineered classical ensembles against state-of-the-art Graph Neural Networks (specifically, the ChemProp Directed Message Passing Neural Network, D-MPNN) using a standardized reference dataset.
- **Empirical Validation of Parsimony over Complexity:** Explicit demonstration that simple classical machine learning ensembles achieve statistical parity ($R^2 \approx 0.75$) with baseline deep learning models reported in recent literature. This establishes that competitive predictive reliability can be attained "out-of-the-box," effectively eliminating the necessity for synthetic data augmentation and computationally expensive 3D geometric preprocessing in low-data regimes.
- **Translational Drug Repurposing Pipeline:** The integration of machine learning predictions with structure-based molecular docking. This hybrid pipeline systematically screens, identifies, and prioritizes FDA-approved compounds, providing a validated computational pathway for rapid therapeutic repurposing.

2. Materials and Methods

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure ??.

2.1. Bioactivity Benchmark Dataset and Curation

This study utilizes a previously published benchmark of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors ?. Adopting the exact baseline dataset compiled by Karimova and Senanayake establishes a rigorous methodological control. This framework isolates the predictive capacity of classical machine learning algorithms from computationally heavy statistical interventions. Omitting synthetic data augmentation during the primary benchmarking phase ensures that performance metrics reflect empirical ground-truth data rather than algorithmic artifacts.

Initial records merged drug-protein interaction parameters and SMILES structures from BindingDB with bioactivity data from ChEMBL ?. Automated standardization protocols, including salt stripping and structural validation, ensured chemical integrity ?. To mitigate experimental noise and adhere to the Organization for Economic Co-operation and Development (OECD) principles for QSAR validation ???, we excluded 117 compounds from the initial 873 entries. This exclusion targeted organometallic complexes—which preclude 3D force-field generation—and molecules with undefined stereocenters that introduce severe conformational ambiguity.

For compounds exhibiting multiple IC_{50} measurements across varying experimental conditions, we applied the Interquartile Range (IQR) method. Duplicate assays with a bioactivity variance exceeding 1.5 log units were discarded as statistical outliers ?. The remaining consistent entries were averaged to yield a single representative measurement per ligand.

This curation protocol yielded a final benchmark of 756 unique small molecules. The continuous target variable was transformed into pIC_{50} ($-\log_{10}(IC_{50})$) ?. This logarithmic conversion standardizes response values and compresses the active range, optimizing the target distribution for robust regression modeling.

To explicitly quantify the performance impact of synthetic data generation reported in the original literature, we implemented an isolated data augmentation protocol exclusively for post-benchmark ablation studies. Following a 15% hold-out partition to prevent data leakage, training features underwent perturbation via additive Gaussian noise. Applying two distinct noise levels ($\sigma = 0.01$ and $\sigma = 0.001$) generated two synthetic variants per original molecule. This procedure effectively triplicated the training space while maintaining absolute isolation of the empirical test set.

2.2. Representation Spaces: Molecular Fingerprints and Conformational Descriptors

We generated four distinct molecular representation spaces using RDKit ? to systematically map the structure-activity relationship. The structural baseline relied on Morgan fingerprints (ECFP4). We computed these with a radius of 2 and folded them into a fixed-length vector of 2048 bits. Extended-connectivity fingerprints offer rapid calculation and capture a vast array of distinct molecular features, including stereochemical topology ?. Classical 2D fingerprint-based models frequently achieve predictive parity with state-of-the-art 3D structure-based models when relying exclusively on ligand information ?.

To establish a physicochemical baseline, the pipeline extracted 217 conventional two-dimensional topological descriptors. Curated 2D molecular representations demonstrate substantial robustness; recent studies confirm they maintain scaffold-independent advantages over 3D alternatives across various property prediction tasks ?. A third hybrid space concatenated these 217 descriptors with the 2048-bit Morgan fingerprints to evaluate potential synergistic effects.

A fourth representation extended the 2D/FP hybrid space with a compact set of ten 3D conformational descriptors (asphericity, eccentricity, inertial shape factor, the first two normalized principal moment ratios, the three principal moments of inertia, radius of gyration, and sphericity index) to assess the incremental predictive value of spatial geometry. The Experimental-Torsion Knowledge Distance Geometry (ETKDGv3) algorithm generated three-dimensional conformers for each molecule ?. Because purely distance-based constraints often distort aromatic rings and sp^2 centers, ETKDGv3 integrates experimental torsion-angle preferences derived from small-molecule crystallographic data. The MMFF94s force field subsequently minimized conformer energy prior to descriptor extraction, and the resulting ten features were concatenated

with the existing 217 2D descriptors and 2048-bit Morgan fingerprints, yielding a combined 2275-feature space.

Although integrating spatial information remains critical for molecular image-based modeling [?] and geometry-augmented representation learning [?], recent empirical evaluations suggest that 3D conformer descriptors yield limited incremental predictive value over curated 2D representations in standard tabular benchmarks [?]. Our nested cross-validation framework explicitly tested this hypothesis. The purely structural Morgan fingerprints yielded a mean R^2 of 0.7417, performing statistically indistinguishably from the computationally expensive 3D hybrid space ($R^2 = 0.7371$, $p = 0.7605$). This empirical parity resolves the representation selection, justifying the exclusive use of the 2048-bit Morgan space for all downstream feature selection, data augmentation, and virtual screening protocols.

2.3. Nested Cross-Validation Protocol and Variance Correction

Evaluating machine learning architectures in low-data regimes demands rigorous isolation between model optimization and performance estimation. Standard cross-validation partitions datasets into k disjoint blocks, systematically rotating them between training and testing phases [?]. While conventional nested cross-validation (NCV) literature frequently defaults to symmetrical 5×5 or 10×10 architectures [???], methodological frameworks do not mandate identical fold counts for inner and outer loops [?]. To guarantee leak-free evaluation, balance computational overhead, and mitigate the model selection bias described in Section 1.3, we implemented a strict 5×5 repeated nested cross-validation protocol for this restricted dataset ($N = 756$). This architecture executes a 5-fold outer loop repeated 5 times, embedding a 5-fold inner loop, mathematically decoupling model assessment from model selection.

Data partitioning strategies dictate the validity of small-scale QSAR models [?]. Bemis-Murcko scaffold splitting is widely advocated to test out-of-domain extrapolation and prevent structural leakage [???]. Applying this technique to a dataset of 756 compounds, however, generates severely imbalanced folds. This imbalance creates artificial “activity cliffs” where entire bioactivity ranges vanish from the training pool, preventing the inner optimization loop from converging on generalizable hyperparameters. Our primary objective is interpolation within the established applicability domain of known TgDHFR inhibitors, rather than extrapolation to entirely novel chemotypes. Random splitting ensures the bioactivity distribution remains statistically consistent across all folds. To empirically validate this partitioning strategy and rule out structural leakage, we evaluated the Tanimoto similarity between training and test sets across all iterations. This standard control confirmed the absence of redundant analog clustering between partitions, ensuring that the reported metrics reflect genuine generalization rather than structural memorization.

The outer loop of the NCV framework evaluates generalization. To achieve lower variance in performance estimation compared to extreme approaches like Leave-One-Out (LOO) [?], we evaluated 25 outer splits. Standard parametric tests assume independent samples. Repeated cross-validation violates this assumption via overlapping training sets across iterations. To neutralize this intra-repetition correlation, we applied the Nadeau and Bengio variance correction [?] across all downstream statistical comparisons. In each iteration, one fold is strictly held out as the test set while the remaining pool passes into the inner loop. This absolute isolation ensures the test data remains hidden during feature and model selection, yielding a truly unbiased out-of-sample prediction estimate [?].

The inner loop governed architecture selection and model optimization. Within each outer training pool, an embedded 5-fold inner loop evaluated all single, pairwise, and three-way stacking combinations (up to 3 base estimators). These combinations were drawn from six diverse regressor families: Random Forest (RF), Extra Trees (ET), Light Gradient Boosting Machine (LightGBM), eXtreme Gradient Boosting (XGBoost), Support Vector Regression (SVR with RBF kernel, $C = 10$), and k -Nearest Neighbors (kNN with cosine distance). This inner loop serves as a structural safeguard against preprocessing leakage. Executing global transformations—such as z-score standardization, median imputation, or supervised feature selection—prior to fold partitioning inadvertently leaks dataset-wide statistics into the training phase [?]. To eliminate this vulnerability, all transformation parameters were strictly computed within each inner training fold and subsequently applied to the validation and test folds without parameter re-estimation [?]. For multi-model combinations, a **StackingRegressor** utilizing **RidgeCV** as the meta-estimator coordinated the meta-learning phase. By confining model selection, descriptor scaling (**StandardScaler**), median imputation, and positive-importance feature selection entirely within these inner partitions, we prevent inadvertent information

leakage. This strict encapsulation guarantees that reported metrics reflect genuine generalizability rather than selection artifacts ???.

2.4. Deep learning baseline: Directed message passing neural networks

Benchmarking classical machine learning ensembles against advanced representation learning required implementing a Directed Message Passing Neural Network (D-MPNN) via the Chemprop framework (version 2.0) ??. To guarantee methodological parity, the D-MPNN underwent the exact 25-fold cross-validation protocol utilized for the classical baselines. Subjecting deep learning architectures to rigorous resampling incurs substantial computational overhead. This strict alignment guarantees performance metrics remain directly comparable and free from single-split variance artifacts.

The architecture processes molecular SMILES strings by mapping them into graph representations, designating atoms as vertices and bonds as edges ?. Initial feature vectors capturing atomic identity and topological connectivity were constructed using `SimpleMoleculeMolGraphFeaturizer()`. The network applied `BondMessagePassing()` to extract embeddings directly from directed edges rather than vertices. This edge-centric design eliminates spurious feedback loops, reducing noise in the learned molecular representations ?. A `MeanAggregation()` function pooled these atomic embeddings into a unified molecular vector.

Prior to training, the continuous pIC_{50} targets required internal normalization. The aggregated embeddings fed into a `RegressionFFN()` equipped with batch normalization to yield the final predictive output. Model weights were optimized via PyTorch Lightning over a maximum of 100 epochs. Unlike the classical ensembles, which underwent extensive hyperparameter optimization, the D-MPNN utilized default Chemprop hyperparameters (e.g., hidden size, depth, and learning rate). This methodological asymmetry stems from both computational constraints and the necessity for structural regularization. Quantitatively, the classical nested cross-validation evaluated 205 independent model fits per outer fold (41 architecture combinations across 5 inner splits), executing in approximately 550 seconds on CPU infrastructure (averaging < 3 seconds per fit). Replicating this combinatorial grid search for the D-MPNN—where a single 100-epoch training cycle demands substantially higher processing time—would inflate the computational overhead by orders of magnitude, rendering it prohibitive for standard screening environments. Beyond hardware limitations, tuning overparameterized neural networks in low-data regimes frequently induces catastrophic memorization rather than genuine generalization ??. Deploying the D-MPNN with default parameters therefore acts as a deliberate regularization strategy. It isolates the intrinsic inductive bias of the graph architecture from the confounding artifacts of small-sample overfitting, establishing a rigorous out-of-the-box deep learning baseline.

The internal loss function optimized during training was Root Mean Square Error (RMSE). Out-of-sample performance was quantified using the coefficient of determination (R^2) and Mean Absolute Error (MAE) averaged across all 25 folds. This rigorous statistical protocol yielded a final cross-validated test performance of $R^2 = 0.6802 \pm 0.0595$ and an MAE of 0.5353, establishing the deep learning baseline for subsequent comparative analysis.

2.5. Virtual Screening Protocol and Ligand Efficiency Profiling

An automated virtual screening pipeline, driven by the highest-performing classical ensemble (Morgan, nested CV $R^2 = 0.7417$), systematically prioritized viable lead candidates from an FDA-approved library ($N = 3,378$). Rather than relying solely on raw predicted bioactivity, the evaluation framework integrated strict applicability domain enforcement with multi-objective ligand efficiency profiling. This approach mitigates the risk of prioritizing “molecularly obese” compounds that exhibit high theoretical potency but possess unfavorable physicochemical properties for downstream development ?.

Initial structural pre-filtering restricted the FDA library to compounds matching the exact elemental composition of the training set (H, C, N, O, F, P, S, Cl, Br, I). This constraint prevents the model from extrapolating into unsupported chemical spaces. A strict physicochemical ceiling (molecular weight ≤ 1000 Da) controlled for collinearity, preventing molecular size from artificially inflating potency predictions.

To enforce the applicability domain quantitatively, an `IsolationForest` algorithm (5% contamination threshold) trained on the original Morgan fingerprint matrix evaluated the remaining compounds. This unsupervised anomaly detection excluded structural outliers, ensuring all advanced candidates resided

strictly within the interpolated chemical space of the known bioactivity distribution. The ensemble's predictive accuracy within this domain was internally validated by accurately recovering the pIC_{50} values of known reference inhibitors, such as Pyrimethamine and Trimethoprim. Ultimately, this hierarchical funnel successfully reduced the initial library of 3,378 compounds to 19 prioritized candidates: the 10 top-ranked novel compounds by predicted potency, plus 9 additional candidates carried forward for orthogonal docking validation, including the established reference inhibitors pyrimethamine and trimethoprim, for downstream orthogonal validation.

Candidate prioritization subsequently incorporated four distinct efficiency metrics to correct *in vitro* activity for physicochemical load. Ligand Efficiency (LE) quantified the binding energy per heavy atom count (HAC), requiring candidates to demonstrate optimal atomic utilization ?:

$$LE = \frac{1.37 \times pIC_{50}}{HAC} \quad (1)$$

To account for heteroatom contributions and total mass, the Binding Efficiency Index (BEI) substituted HAC with molecular weight (MW) ?:

$$BEI = \frac{pIC_{50}}{MW} \times 1000 \quad (2)$$

Furthermore, to balance potency against lipophilicity and mitigate the risk of nonspecific binding, Lipophilic Ligand Efficiency (LLE) was computed as the difference between predicted potency and the calculated partition coefficient ($LLE = pIC_{50} - cLogP$). Finally, the Surface Efficiency Index (SEI) evaluated potency relative to the topological polar surface area (TPSA), providing a proxy for membrane permeability potential ($SEI = (pIC_{50}/TPSA) \times 100$). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts ?, these four metrics (LE, BEI, LLE, SEI) operated strictly as secondary multi-objective ranking filters rather than primary objective functions.

2.6. Orthogonal Physics-Based Validation via Molecular Docking

Orthogonal structure-based evaluation via AutoDock Vina v1.2.0 ?? complemented the ligand-based predictions. This physical validation translates abstract chemical representations into spatial coordinates, mitigating the risk of false positives induced by algorithmic scaffold memorization. By ensuring genuine structural complementarity, the docking protocol acts as a thermodynamic filter. Vina employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations ?.

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG, chain B) ??. To accurately simulate the native biological environment, the NADPH cofactor was retained within the catalytic pocket. Following the removal of crystallographic water and local minimization, receptor protonation states were assigned at a physiological pH of 7.4. Polar hydrogens and Gasteiger charges were subsequently computed using AutoDockTools to generate the required PDBQT format ?. The Vina search space was centered on the native CP6 ligand coordinates ($x = 3.689, y = 39.992, z = -62.818$) with grid box dimensions set to $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32. This configuration permitted unrestricted exploration of ligand orientations and flexible rotatable bonds, thereby preventing crystal-pose centering bias ?.

Prior to screening the machine learning candidates, the docking protocol underwent rigorous validation via re-docking of the co-crystallized CP6 ligand into the TgDHFR active site. The protocol successfully reproduced the native crystallographic binding mode, yielding a top-ranked pose with a centroid distance of 0.32 Å. This deviation falls well below the universally accepted 2.0 Å RMSD threshold, confirming the structural fidelity of the grid parameters and the reliability of the scoring function for this specific target.

Following validation, Vina evaluated the 3D conformers of the 19 top-ranked FDA candidates alongside the reference controls. Poses were ranked based on the predicted Gibbs free energy of binding (ΔG) in kcal/mol, where lower energies denote stronger thermodynamic stability ?. To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency

($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criteria were flagged as potential ML false positives. Finally, to quantify the monotonic agreement between the two orthogonal methods without assuming linearity, a Spearman rank correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores ($\rho = -0.369$, $p = 0.1203$).

2.7. In Silico Pharmacokinetic (ADMET) Profiling

Following consensus validation, a comprehensive predictive pharmacokinetic (ADMET) assessment evaluated the surviving candidates to reduce late-stage attrition risks. Candidates first underwent evaluation against Lipinski’s Rule of Five criteria, strictly eliminating molecules exceeding 5 hydrogen bond donors, 10 hydrogen bond acceptors, or a calculated LogP of 5. To assess critical toxicity and absorption endpoints, custom machine learning models were trained using curated datasets from the Therapeutics Data Commons (TDC). Molecular structures were encoded as 2048-bit Morgan fingerprints (radius 2) and processed through Random Forest architectures comprising 100 estimators.

A `RandomForestClassifier` trained on the TDC hERG dataset evaluated cardiac toxicity risk. This served as a strict safety filter, advancing only compounds with a low probability of hERG channel blockade ($P < 0.5$). Unlike this destructive toxicity threshold, a `RandomForestRegressor` trained on the Caco2-Wang dataset utilized intestinal absorption predictions to categorize the optimal administration route. Compounds demonstrating moderate-to-high permeability ($\log P_{app} > -5.15$) were classified as orally compliant. Conversely, candidates falling below this threshold were not discarded; they were retained and flagged as viable parenteral therapeutics. This non-destructive pharmacokinetic profiling ensured that the prioritized consensus hits represent structurally validated, safe starting points for therapeutic repurposing, accommodating both oral and intravenous delivery modalities.

2.8. Computational Environment and Reproducibility

To guarantee algorithmic transparency and absolute computational reproducibility, all data curation, model training, and evaluation pipelines were programmatically implemented in Python (version 3.11.16). The computational environment relied on a standardized core software stack:

- **Cheminformatics and Featurization:** RDKit (version 2025.9.3) executed all molecular parsing, structural standardization, and descriptor generation (2D, 3D, and ECFP4).
- **Classical Machine Learning:** The nested cross-validation framework and baseline algorithms leveraged `scikit-learn` (version 1.8.0), `xgboost` (version 3.1.2), and `lightgbm` (version 4.6.0). Data manipulation and matrix operations were managed via `pandas` (version 2.3.3) and `numpy` (version 2.3.5).
- **Deep Learning Infrastructure:** The D-MPNN graph models utilized `chemprop` (version 2.0) and `lightning`, with underlying tensor operations supported by PyTorch (`torch` $\geq 2.0.0$).
- **Hardware Specifications:** All model training, hyperparameter tuning, and evaluation protocols ran on an Ubuntu 26.04 LTS workstation equipped with an Intel(R) Xeon(R) CPU E5-2695. The deliberate use of CPU-only acceleration explicitly benchmarks the computational accessibility and hardware demands of both classical and deep learning architectures in low-resource settings.

3. Results

3.1. Comparative Performance of Classical ML Across Feature Spaces

We established a rigorous baseline for TgDHFR inhibitor prediction by evaluating classical machine learning ensembles across four molecular representations using a 25-fold nested cross-validation framework. The purely structural representation utilizing Morgan fingerprints (ECFP4, 2048 bits) yielded the highest predictive performance ($R^2 = 0.7417 \pm 0.0493$, $MAE = 0.4841$). To definitively rule out chance correlation, we subjected this optimized baseline to a 100-iteration y-randomization protocol. The permuted models collapsed to a mean R^2 of -0.0147 ± 0.0094 , generating a massive statistical separation from the true

Table 1

Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2 (mean $\pm$ std)
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.7417 $\pm$ 0.0493
Baseline ?	2D/3D/FP	None	0.75 ^a
Augmented Final Model ?	2D/3D/FP (Selected)	Gaussian Noise	0.85 ^a

^a Standard deviation not reported (NR) in the original study.

predictive baseline ($Z = 80.0$). This confirms the ensemble maps genuine physicochemical relationships rather than memorizing noise.

Figure ?? illustrates the distribution of explained variance across all evaluated feature spaces. Within the Morgan fingerprint space, architectural selection exhibited severe fluidity. Rather than converging on a single dominant hierarchy, five distinct architectures tied as the most frequently selected configurations (12.0% each, 3/25 folds): LGBM+XGB+SVM, RF+SVM, RF+XGB+SVM, SVM, and XGB+SVM+kNN. This distribution confirms that while aggregated predictive performance remains robust across the outer folds, the inherent data sparsity of the 756-compound benchmark prevents the inner optimization loop from isolating a universally superior stacking hierarchy.

Escalating feature complexity failed to yield commensurate predictive gains. Integrating RDKit 2D descriptors with Morgan fingerprints (analogous to the original study’s “2D/FP” model) produced an R^2 of 0.7306 ± 0.0609 and an MAE of 0.4901. The most complex representation, incorporating 3D geometric descriptors derived from ETKDGV3 embedding and MMFF94s optimization (analogous to the “2D/3D/FP” model), attained an R^2 of 0.7371 ± 0.0610 and an MAE of 0.4828. A paired t-test comparing the purely structural Morgan fingerprint ensemble against the highly engineered 2D/3D/FP model revealed no statistically significant difference in explained variance ($t = 0.308$, $p = 0.7605$). To rigorously assess error magnitude differences while avoiding pseudoreplication artifacts, we applied a Nadeau-Bengio corrected paired t-test to the cross-validated MAE distributions. This confirmed the variance in absolute error lacks statistical significance ($t = 0.098$, $p = 0.9230$). These results indicate that conformational 3D featurization offers no measurable predictive advantage for this specific benchmark. Conversely, relying exclusively on 2D physicochemical descriptors severely degraded performance ($R^2 = 0.6796 \pm 0.0658$, MAE = 0.5300). Topological fingerprints act as the primary drivers of predictive capacity.

We benchmarked our highest-performing classical model against the incremental progression reported in the original study to contextualize the computational cost-benefit ratio of advanced statistical techniques (Table ??). Our classical stacking ensemble trained solely on Morgan features ($R^2 = 0.7417$) performed comparably to their unselected 2D/3D/FP baseline model ($R^2 = 0.75$). Evaluated under a Nadeau-Bengio corrected one-sample t-test, this variance lacks statistical significance ($t = -0.315$, $p = 0.7558$). Furthermore, our baseline 95% bootstrap confidence interval [0.7275, 0.8624] strictly encompasses the literature baseline, confirming statistical parity.

However, the literature model utilizing permutation importance-based feature selection ($R^2 = 0.82$) represents a statistically significant improvement over our unselected baseline ($t = -2.953$, $p = 0.0069$), attributable to the supervised isolation of pharmacophoric signals. Similarly, the original study’s final data-augmented deep neural network ensemble ($R^2 = 0.85$) resides outside our baseline 95% CI, representing a statistically significant deviation ($t = -4.084$, $p = 0.0004$). These comparative results isolate the true source of predictive gains in this low-data regime. Achieving a competitive baseline performance ($R^2 \approx 0.75$) does not require computationally expensive 3D descriptors or complex neural network architectures; simple 2D fingerprints paired with classical machine learning algorithms suffice. The performance gaps between our baseline and the original study’s advanced models directly quantify the impact of supervised feature selection and synthetic data augmentation rather than inherent architectural superiority.

3.2. Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph Neural Networks (GNNs) dominate molecular property prediction by extracting representations directly from graph topologies. Their predictive utility, however, degrades sharply under data scarcity. To

evaluate whether deep representation learning offers intrinsic advantages for the TgDHFR dataset (756 compounds), we benchmarked our optimal classical ensemble against a Directed Message Passing Neural Network (D-MPNN) using the Chemprop framework.

The D-MPNN yielded a cross-validated out-of-sample R^2 of 0.6802 ± 0.0595 and a Mean Absolute Error of 0.5353. This network failed to match the classical stacking ensemble operating on pre-computed Morgan fingerprints ($R^2 = 0.7417 \pm 0.0493$). This performance deficit stems directly from data scarcity and methodological asymmetry. The classical models underwent extensive hyperparameter optimization and architecture selection via the nested cross-validation pipeline. In contrast, the D-MPNN was evaluated using baseline hyperparameters without an equivalent exhaustive tuning phase. Restricted to 756 training instances, the unoptimized D-MPNN cannot learn graph convolutions capable of surpassing the predictive baseline established by fixed, topological ECFP4 descriptors.

This convergence plateau provides contextual validation for the methodologies in the original study ?. The poor performance of the unaugmented D-MPNN explains the prior reliance on heavy statistical interventions—specifically, Gaussian data augmentation—to force predictive accuracy ($R^2 = 0.85$, $p < 0.001$). Evaluated on unaugmented, data-limited chemical spaces, architectural complexity does not guarantee superior generalization. Classical tree-based ensembles utilizing fixed molecular descriptors remain the most robust, parsimonious, and computationally efficient alternative. Figure ?? details the comparative performance across all baseline models and the GNN.

3.3. Virtual Screening and Prioritization of FDA-Approved Candidates

Transitioning from theoretical benchmarking to applied therapeutic discovery required deploying the optimized classical ensemble (Morgan, $R^2 = 0.7417$) against a curated library of FDA-approved compounds. Prior to evaluating uncharacterized chemical spaces, the pipeline’s predictive reliability underwent quantification using established TgDHFR inhibitors retained within the dataset. The algorithm recovered these internal controls with high fidelity. It assigned a predicted pIC_{50} of 6.3175 to pyrimethamine (experimental: 6.56, absolute error: 0.2425) and 5.6325 to trimethoprim (experimental: 5.57, absolute error: 0.0625). This empirical alignment confirms the ensemble’s capacity to interpolate bioactivity accurately within its defined applicability domain. To visualize this chemical space alignment, Principal Component Analysis (PCA) of the Morgan fingerprints confirmed that the prioritized FDA candidates fall strictly within the dense regions of the training distribution (Figure S1).

Following unsupervised Isolation Forest filtering to exclude structural outliers, the virtual screening protocol prioritized candidates based on absolute predicted potency and multi-objective ligand efficiency. Table ?? delineates the top ten prioritized compounds. Classic antifolate analogs—specifically methotrexate, trimetrexate, and folic acid—dominated the upper echelon of absolute affinity, all exceeding a predicted pIC_{50} of 7.0. Absolute potency, however, frequently scales with molecular weight. This size dependency can artificially inflate binding scores and obscure the intrinsic thermodynamic quality of the underlying scaffold.

Correcting for physicochemical load via heavy-atom and molecular weight penalties revealed a distinct hierarchy of efficient binders. Smaller architectures demonstrated vastly superior atomic utilization. Pyrimethamine (LE = 0.51, BEI = 25.40), the potassium-sparing diuretic triamterene (LE = 0.47, BEI = 25.87), and the phenothiazine derivative methylene blue (LE = 0.45, BEI = 21.61) emerged as the most structurally optimized candidates. Triamterene and methylene blue provide particularly compelling repurposing profiles. They achieve robust predicted bioactivities while maintaining exceptional ligand efficiency indices, indicating highly complementary interactions with the TgDHFR catalytic pocket relative to their minimal steric footprints.

3.4. Integrated ML-docking consensus filtering and false-positive mitigation

Standalone virtual screening methodologies generate false-positive artifacts. To mitigate this vulnerability, we implemented an orthogonal consensus filtering pipeline integrating ligand-based machine learning predictions with physics-based molecular docking. Spearman’s rank correlation coefficient evaluated the global agreement between ML-predicted pIC_{50} values and AutoDock Vina binding energies across the prioritized subset ($N = 19$). This non-parametric approach ($\rho = -0.369$, $p = 0.1203$) handles restricted sample sizes and isolates monotonic trends without assuming linear dependencies between topological bioactivity interpolation and thermodynamic scoring.

Table 2

Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Methotrexate	7.5470	0.31	16.61	7.28	3.58
2	Trimetrexate Glucuronate (CID 336805) ^a	7.3500	0.25	13.04	7.90	2.91
3	Trimetrexate Glucuronate (CID 54949) ^a	7.3500	0.25	13.04	7.90	2.91
4	Pralatrexate	7.1884	0.28	15.05	6.21	3.47
5	Folic Acid	7.1448	0.31	16.19	7.19	3.35
6	Bendamustine Hydrochloride	6.9405	0.40	17.58	3.25	11.89
7	Methylene Blue	6.9123	0.45	21.61	7.41	36.11
8	Trifarotene	6.8626	0.28	14.93	0.88	9.80
9	Fentanyl Citrate	6.8592	0.25	12.98	3.97	4.41
10	Chlorambucil	6.8371	0.49	22.47	3.46	16.87

^a Duplicate PubChem CID entries mapping to the same underlying structure in the FDA library.

The negative correlation confirms that higher predicted potencies trend with stronger binding energies. The absence of statistical significance ($p > 0.05$) validates the methodological design: two-dimensional ligand-based machine learning and three-dimensional physics-based docking capture fundamentally orthogonal aspects of molecular interaction. The ML ensemble interpolates bioactivity from topological patterns within the training distribution, whereas Vina calculates thermodynamic stability via spatial complementarity and electrostatic interactions. Relying on either method in isolation exposes the screening process to distinct classes of false positives.

The consensus approach identified and eliminated these methodological artifacts. Bisacodyl perfectly illustrates a physics-based false positive. It exhibited strong thermodynamic binding energies during structural validation ($\Delta G = -8.76$ kcal/mol), driven by the spurious accommodation of its flexible, hydrophobic scaffold within the binding pocket. The classical ML model, however, correctly assigned a low predicted affinity ($pIC_{50} = 5.3229$) to this compound, vetoing its progression. Conversely, the ML model generated exactly 3 false positives—methylene blue, bendamustine hydrochloride, and chlorambucil—each exceeding the ML potency threshold ($pIC_{50} > 6.5$) yet failing to satisfy the stringent docking energy criterion ($\Delta G > -8.0$ kcal/mol). Dual validation thus prevents the misallocation of experimental resources toward single-method screening artifacts.

Orthogonal docking isolated exactly 8 consensus hits from the 19 evaluated compounds that satisfied both the ML potency threshold ($pIC_{50} > 6.5$) and the stringent docking energy criteria ($\Delta G < -8.0$ kcal/mol). These validated hits are pralatrexate ($\Delta G = -9.50$ kcal/mol), trifarotene ($\Delta G = -9.01$ kcal/mol), methotrexate ($\Delta G = -8.71$ kcal/mol), fentanyl citrate ($\Delta G = -8.66$ kcal/mol), folic acid ($\Delta G = -8.57$ kcal/mol), trimetrexate glucuronate (CID 336805 and CID 54949, both $\Delta G = -8.48$ kcal/mol), and triamterene ($\Delta G = -8.04$ kcal/mol). The primary consensus candidate, Pralatrexate, demonstrated a strong ML-predicted pIC_{50} of 7.1884 alongside the most favorable docking score of -9.50 kcal/mol.

This lightweight, classical pipeline matches the statistical performance of unaugmented deep learning models while discovering biologically viable, physics-validated inhibitors. Isolating high-confidence repurposing candidates without synthetic data generation or 3D feature engineering yields a superior computational cost-benefit ratio for practical drug discovery in low-data regimes.

4. Discussion

4.1. The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph Neural Networks theoretically offer superior representational capacity by learning molecular features directly from graph topology, thereby bypassing the need for manual feature engineering. However, this architectural flexibility demands extensive training data to converge on generalizable chemical representations. In the context of the TgDHFR dataset, which comprises only 756 compounds, the GNN architecture failed to leverage its theoretical superiority, plateauing at a predictive performance statistically indistinguishable from classical machine learning approaches.

This convergence failure highlights the critical role of explicit inductive bias in low-data regimes. The fundamental lack of explicit chemical priors in deep representation learning forces these architectures to construct a continuous latent space entirely from scratch. With fewer than 1,000 training instances, the resulting latent space density is severely restricted, creating a sparse manifold where interpolation between data points becomes highly unstable. To counteract this sparsity, deep learning frameworks structurally require artificial variance injection. Consequently, achieving high predictive accuracy with neural networks on this specific dataset necessitates extensive Gaussian noise-based data augmentation, as demonstrated in the original study [?], to artificially inflate latent space density and force convergence.

Classical tree-based ensembles operating on pre-computed Morgan fingerprints bypass the need for end-to-end representation learning by utilizing fixed, domain-specific descriptors. These fingerprints encode established chemical heuristics—specifically, circular topological substructures—directly into the feature space. For small-scale datasets, this explicit pre-computation acts as a natural regularizer, rigidly constraining the hypothesis space and preventing the model from memorizing noise. Our results explicitly demonstrate that exploiting the inductive bias inherent to traditional structural fingerprints entirely eliminates the necessity for computationally burdensome data augmentation in this specific context.

Nevertheless, this architectural superiority is strictly regime-dependent. While deterministic feature engineering provides a parsimonious and highly reliable strategy for the current low-data regime ($N = 756$), this advantage is expected to invert as the dataset scales. In high-data environments ($N \gg 1000$), the rigid constraints of pre-computed fingerprints typically become an informational bottleneck. Under those conditions, the flexible representational capacity of GNNs allows them to learn task-specific embeddings that systematically outperform classical descriptors.

4.2. Dimensionality, Conformational Overhead, and Model Efficiency

Computational efficiency determines the practical viability of molecular representations in high-throughput virtual screening. Generating 3D geometric descriptors requires extensive conformational sampling and energy minimization (e.g., ETKDGV3 embedding and MMFF94s optimization). This conformational overhead scales non-linearly, creating severe processing bottlenecks when screening libraries expand into the millions of compounds.

Empirical results demonstrate that this computational expenditure provides no predictive advantage for TgDHFR inhibition. The composite 2D/3D/FP representation yielded an R^2 of 0.7371, which remains statistically indistinguishable from the purely topological Morgan fingerprint model ($R^2 = 0.7417$; Nadeau-Bengio corrected $p = 0.9230$). This parity confirms that 2D connectivity matrices capture the requisite pharmacophoric variance.

Parsimony therefore drives the optimal screening strategy. Excluding 3D descriptors and synthetic data augmentation eliminates processing latency during both feature extraction and model training. For this specific target, high-dimensional representations introduce computational friction without improving model fidelity. Consequently, lightweight 2D fingerprint ensembles offer a strictly superior, highly scalable architecture for virtual screening under constrained data conditions.

4.3. Methodological Scope, Limitations, and Translational Potential

The predictive reliability of any quantitative structure-activity relationship (QSAR) model is inherently bounded by the chemical diversity of its training corpus. Given the dataset size of 756 compounds, the applicability domain of our optimized ensemble is strictly restricted to the structural scaffolds and physicochemical properties represented within the curated TgDHFR data. Consequently, extrapolating predictions to highly divergent chemotypes or novel synthetic spaces carries an increased risk of reduced predictive confidence.

A specific vulnerability intrinsic to 2D circular fingerprints in low-data regimes is their susceptibility to activity cliffs—pairs of structurally similar molecules that exhibit radically divergent bioactivities due to subtle chemical modifications. A single bioisosteric substitution or methylation can alter a critical hydrogen bond or steric fit within the binding pocket. Because Morgan fingerprints map topological neighborhoods into bit vectors without explicitly encoding 3D spatial conformation or electrostatic field potentials, they risk mapping active and inactive cliff-forming analogs to virtually identical representations. We strategically mitigated this limitation through our downstream multi-tier architecture. The orthogonal integration of

physics-based molecular docking explicitly evaluates 3D spatial complementarity and steric clashes, acting as a structural filter that detects activity cliffs otherwise invisible to pure 2D topological models.

An additional methodological constraint involves the extensive architecture search executed within the inner optimization loop. Evaluating up to 41 distinct stacking combinations across six regressor families introduces a latent risk of model selection bias. While the nested cross-validation framework mathematically isolates this optimization phase from the outer test folds—thereby mitigating spurious performance inflation—it does not entirely eliminate architectural instability. To transparently quantify this variance, Supplementary Table S1 details the selection frequency of each meta-ensemble across the 25 outer folds. Within the optimal Morgan fingerprint space, the LGBM+XGB+SVM configuration emerged as the most stable architecture, yet it was selected in only 12.0% of the iterations. This distribution confirms that while the nested design successfully prevents test-set leakage, the inherent data sparsity of the 756-compound benchmark inevitably restricts the convergence of the inner loop onto a single, universally dominant stacking hierarchy.

Despite these boundary conditions, the programmatic pipeline developed in this study exhibits high transferability to other neglected tropical diseases (NTDs) and orphan targets. Because the framework relies on classical ensemble learning rather than data-hungry deep neural networks, it maintains predictive stability in low-data regimes (e.g., datasets with fewer than 1,000 compounds). This architecture provides a computationally lightweight triage mechanism that can be rapidly deployed for other under-researched pathogens. By systematically filtering extensive libraries of approved drugs to isolate a tractable subset of candidates, the pipeline minimizes the computational overhead required for subsequent physics-based simulations and reduces the attrition rate in downstream experimental validations.

Finally, the integration of ADMET profiling contextualizes the clinical viability of the prioritized candidates. The top-ranked hybrid hit, Pralatrexate (-9.50 kcal/mol), alongside the reference drug Methotrexate, exhibited low predicted Caco-2 permeability, classifying them as parenteral rather than oral candidates. Rather than a limitation, this pharmacokinetic profile is mechanistically consistent with classical antifolates, which are typically highly polar, structurally complex molecules. Clinically, parenteral administration aligns directly with the treatment protocols for severe, acute manifestations of the disease, such as toxoplasmic encephalitis in immunocompromised patients, where intravenous delivery ensures rapid systemic bioavailability. Consequently, the pipeline not only identifies potent target binders but also accurately anticipates their clinical administration routes, reinforcing its utility in applied pharmacological repurposing.

4.4. Conclusions and Future Directions

This study re-evaluates the methodological prerequisites for molecular property prediction in low-data environments ($N = 756$), yielding three principal findings. First, classical machine learning ensembles operating on 2D topological fingerprints provide sufficient explicit inductive bias ($R^2 = 0.7417 \pm 0.0493$) to surpass the predictive utility of unaugmented deep learning architectures ($R^2 = 0.6802$) and computationally expensive 3D descriptors. This advantage, however, remains strictly bounded to low-data regimes where complex neural architectures suffer from parameter underdetermination. Second, rigorous nested cross-validation remains imperative to eliminate optimistic performance bias and accurately quantify out-of-sample generalization.

Third, integrating ligand-based statistical predictions with physics-based molecular docking and ADMET profiling establishes a robust consensus filter. This orthogonal validation mitigates the false-positive rates inherent to isolated screening methodologies. By enforcing strict pharmacokinetic constraints—specifically evaluating hERG cardiotoxicity and Caco-2 intestinal permeability—the pipeline isolates 8 structurally and translationally viable candidates. Most notably, Pralatrexate emerges as a high-priority candidate for therapeutic repurposing, demonstrating both strong predicted affinity and a compliant safety profile.

Future investigations must transition these computational findings into experimental validation. *In vitro* enzymatic assays are required to confirm the inhibitory kinetics of Pralatrexate and the remaining consensus candidates against TgDHFR. Subsequent *in vivo* efficacy models in *Toxoplasma gondii*-infected cohorts will determine their systemic viability. Additionally, expanding the computational framework to evaluate multi-target selectivity profiles—specifically quantifying the binding affinity differential between TgDHFR and human DHFR—will be necessary to anticipate and minimize off-target toxicity in clinical applications.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following [link](#)

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

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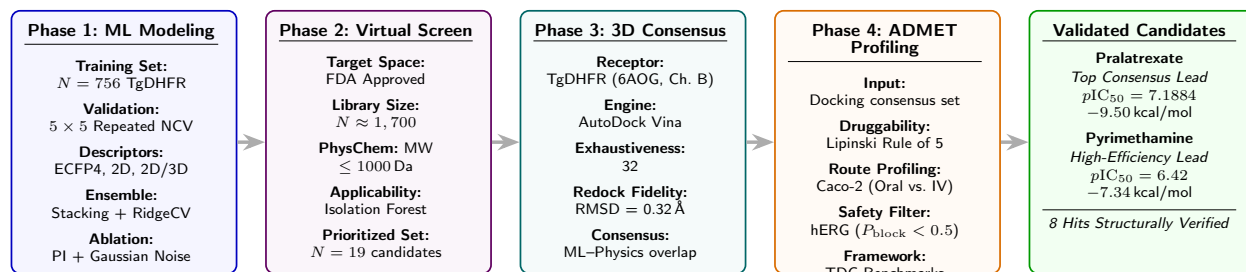


Figure 1: Schematic overview of the multi-tier computational repurposing and validation pipeline. The hierarchical funnel integrates: (1) 2D ligand-based representation benchmarking under nested cross-validation, (2) applicability domain-constrained screening of FDA-approved compounds, (3) orthogonal physics-based consensus validation via molecular docking (AutoDock Vina), and (4) in silico ADMET profiling (Caco-2 permeability route categorization and hERG cardiotoxicity filtering) to isolate clinically viable TgDHFR candidates.

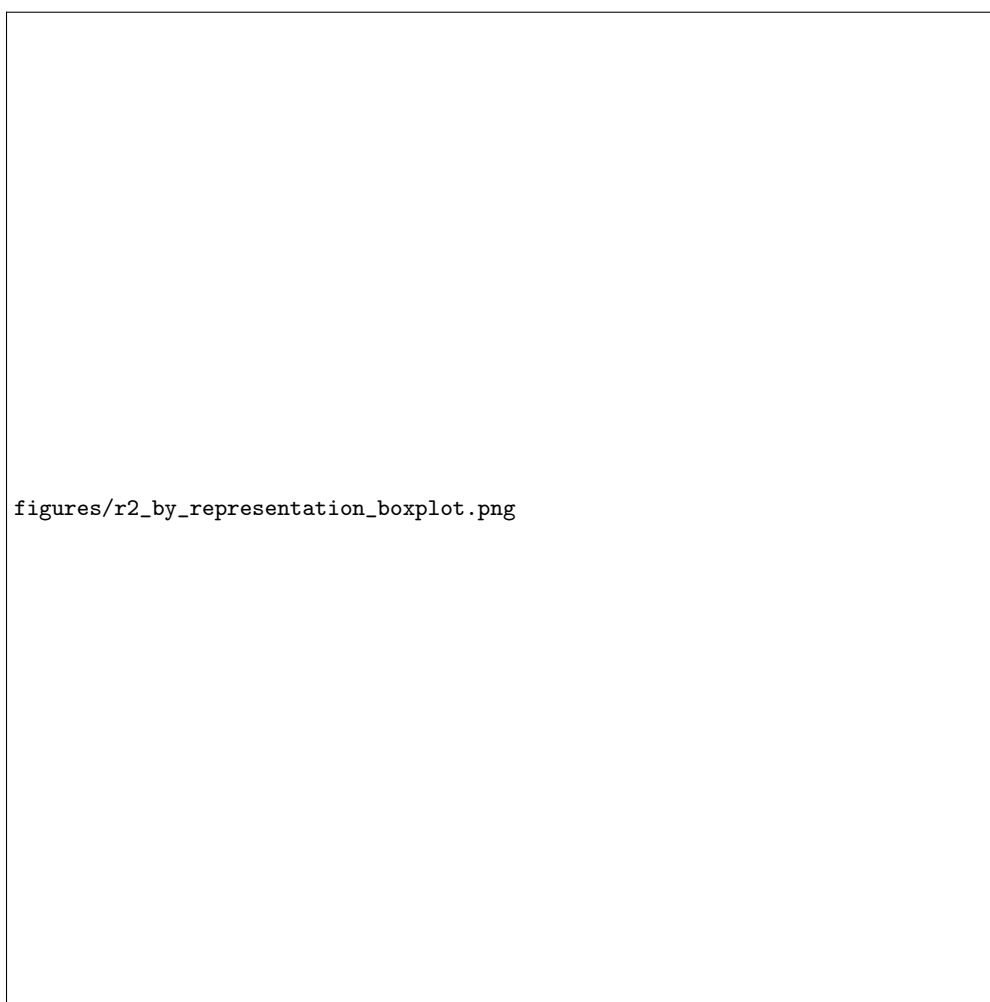


Figure 2: Nested cross-validation R^2 distributions across four molecular representations. The purely structural Morgan fingerprint representation demonstrates equivalent predictive capacity to highly engineered 3D descriptors, establishing a robust classical baseline.

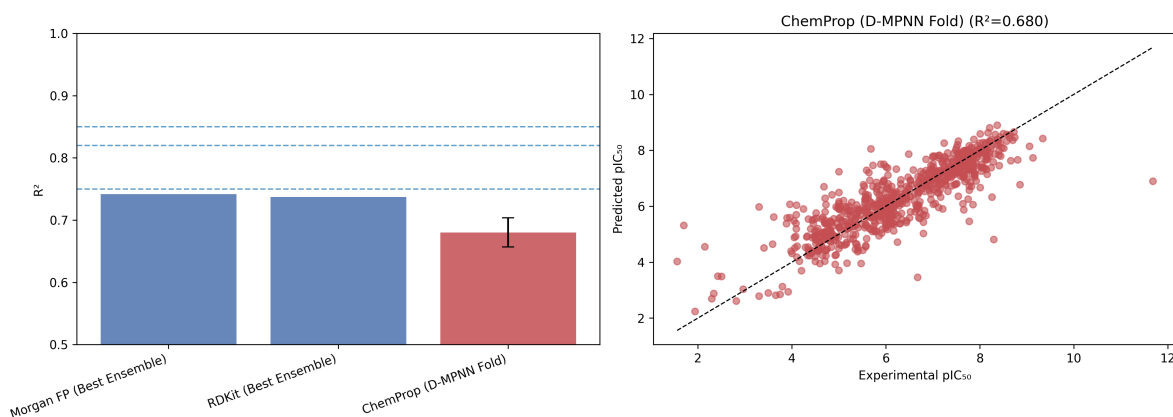


Figure 3: Comparative predictive performance of classical machine learning ensembles, the Chemprop D-MPNN, and the original study's baseline and final models. The scatter plot details the predicted versus experimental pIC_{50} values for the D-MPNN.